

Program 6 : Deploy an Automated Build Pipeline Using Docker hub

Project Structure

```
.
├── build
├── dist
├── Dockerfile
├── node_modules
├── package.json
├── package-lock.json
├── README.md
└── src
```

Step 1: Create Github PAT

Github -> settings -> developer settings -> personal access tokens
choose classic tokens

generate with: write: package and read:package permissions
save token safely

Step 2: Create docker file and build Image

docker build -t program6 .

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$ docker build -t lab6 .
[+] Building 14.0s (15/15) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 423B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  3.6s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:3960ed74dfe320a67bf8da9555b6bade25ebda2b22b608  4.5s
=> => resolve docker.io/library/node:20-alpine@sha256:3960ed74dfe320a67bf8da9555b6bade25ebda2b22b6081d2f60fd  0.0s
=> => sha256:c2b4197efb6ccd7f8b482ae7800f1c9c78c044ea192587887300080bcff6b2c9  42.78MB / 42.78MB  2.5s
=> => sha256:3dcec91425079e7b455efc5f2a18d026450c47c9382c41897620afc6b1424e44  1.26MB / 1.26MB  0.8s
=> => sha256:41b3afaea3b1b1ab04a268431e10dcace7883019a5da7d326aa35dc9713fcb5  445B / 445B  1.1s
=> => sha256:3960ed74dfe320a67bf8da9555b6bade25ebda2b22b6081d2f60fd7d5d430e9c  7.67kB / 7.67kB  0.0s
=> => sha256:2dd4425a1528b0c47e065c88b3d4b73830c8fe68132726cd8485453059774468  1.72kB / 1.72kB  0.0s
=> => sha256:1b3a22d17cb67849eb8a0f8c775b3287b6d8616f1f41ad227658ff6d0c8167c3  6.52kB / 6.52kB  0.0s
=> => extracting sha256:c2b4197efb6ccd7f8b482ae7800f1c9c78c044ea192587887300080bcff6b2c9  1.4s
=> => extracting sha256:3dcec91425079e7b455efc5f2a18d026450c47c9382c41897620afc6b1424e44  0.1s
=> => extracting sha256:41b3afaea3b1b1ab04a268431e10dcace7883019a5da7d326aa35dc9713fcb5  0.0s
=> [internal] load build context                                  0.4s
=> => transferring context: 1.12MB                                   0.3s
=> [builder 2/6] WORKDIR /app                                    0.5s
=> [builder 3/6] COPY package.json package-lock.json ./         0.1s
=> [builder 4/6] RUN npm install                                 2.3s
=> [builder 5/6] COPY . .                                        1.4s
=> [builder 6/6] RUN npm run build                               0.6s
=> [stage-1 3/6] COPY --from=builder /app/package.json ./       0.1s
=> [stage-1 4/6] COPY --from=builder /app/package-lock.json ./  0.1s
=> [stage-1 5/6] COPY --from=builder /app/dist ./dist           0.1s
=> [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules 0.3s
=> exporting to image                                           0.2s
=> => exporting layers                                           0.1s
=> => writing image sha256:2356410e1a3c7bfcba1931b7eee764edf6ddec0495a38420548879b518850c8  0.0s
=> => naming to docker.io/library/lab6                           0.0s
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$
```

Step 3: Tag image for Docker Hub

```
docker tag lab6 <dockerhub_username>/lab6
```

Step 4: Push image to docker hub

```
docker login -u <docker_username>
```

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$ docker login -u jayeshjain11
WARNING: Error parsing config file (/home/jayesh-jain/.docker/config.json): invalid character '#' looking for beginning of object key string

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:

WARNING! Your credentials are stored unencrypted in '/home/jayesh-jain/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded

1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$ docker push jayeshjain11/lab6:latest
The push refers to repository [docker.io/jayeshjain11/lab6]
8dfd0c4d162b: Pushed
ab16e98f4743: Pushed
cd7c716c6e74: Pushed
647b492c25a2: Pushed
32fe6c917add: Pushed
919c27cad7cf: Mounted from library/node
4e05d5f29550: Mounted from library/node
bc2e73ec7de4: Mounted from library/node
7bb20cf5ef67: Mounted from jayeshjain11/frontend
latest: digest: sha256:a3eab48834d29706c978cefc0648f22facb7b2a8b3d30c68ed9ffdc306fd74cd size: 2196
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$
```

Step 5 : Tag image for github container registry (GHCR)

```
docker tag program6 ghcr.io/<github_username>/program6:latest
```

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$ echo "ghp_cK1PRI5fPQ7Z4pyt9SxB3SRUKq0qR73HSb3b" | docker login ghcr.io -u jayesh-jain-11 --password-stdin

WARNING! Your credentials are stored unencrypted in '/home/jayesh-jain/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$
```

Step 6: Push Image to GHCR

```
docker push ghcr.io/<githubusername>/program-6:latest
```

```
Login Succeeded
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$ docker push ghcr.io/jayesh-jain-11/lab6
Using default tag: latest
The push refers to repository [ghcr.io/jayesh-jain-11/lab6]
8dfd0c4d162b: Pushed
ab16e98f4743: Pushed
cd7c716c6e74: Pushed
647b492c25a2: Pushed
32fe6c917add: Pushed
919c27cad7cf: Pushed
4e05d5f29550: Pushed
bc2e73ec7de4: Pushed
7bb20cf5ef67: Pushed
latest: digest: sha256:a3eab48834d29706c978cefc0648f22facb7b2a8b3d30c68ed9ffdc306fd74cd size: 2196
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$
```

Step 7: Connect Project to Github

```
git init
git remote add origin https://github.com/<github-username>/program-6.git
git remote -v
```

Step 8: Push source code to github

```
git add .
git commit -m "Initial commit"
```

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$ git remote -v
origin  https://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@github.com/jayesh-jain-11/devops_lab_6 (fetch)
origin  https://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@github.com/jayesh-jain-11/devops_lab_6 (push)
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab6$
```

Step 9 : Verify Uploads

Github

verify github repo

check packages : image should appear as ghcr.io/<github-username>/lab6:latest

The screenshot shows the GitHub profile of Jayesh Kumar R (username: jayesh-jain-11). The profile includes a circular avatar with a green and white geometric pattern. Below the avatar, the name "Jayesh Kumar R" and username "jayesh-jain-11" are displayed, along with an "Edit profile" button. To the right, there is a search bar and filters for "Type: All", "Visibility: All", and "Sort by: Most downloads". Below these, a section titled "2 packages" lists two packages:

- lab_6** (Private): Published on Dec 12, 2025 by Jayesh Kumar R. Download count: 0.
- lab6** (Private): Published on Dec 11, 2025 by Jayesh Kumar R in [jayesh-jain-11/devops_lab6](#). Download count: 0.

Step 10 : Verify upload in Docker Hub

The screenshot shows the Docker Hub "Repositories" page for the namespace "jayeshjain11". The page header indicates "All repositories within the jayeshjain11 namespace." Below this, there is a search bar "Search by repository name" and a dropdown menu "All content". A blue button "Create a repository" is located on the right. The main content is a table with the following columns: "Name", "Last Pushed", "Contains", "Visibility", and "Scout".

Name	Last Pushed	Contains	Visibility	Scout
jayeshjain11/lab6	about 17 hours ago	IMAGE	Public	Inactive